10.5.1

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Question

Let ABC be a right triangle in which $AB = 6cm, BC = 8cm$ and $\angle B = 90$. BD is the perpendicular from **B** on AC. The circle through $\mathbf{B}, \mathbf{C}, \mathbf{D}$ is drawn. Construct the tangents from **A** to this circle.

Theoretical Solution

Let,

$$\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 0 \\ 6 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} 8 \\ 0 \end{pmatrix} \tag{1}$$

Let the equation of the line passing through $\mathbf{A}$ and $\mathbf{C}$ be,

$$\mathbf{n}^\top \mathbf{x} = c \tag{2}$$

The direction vector of line AC is given by,

$$\mathbf{m} = \mathbf{A} - \mathbf{C} \equiv \begin{pmatrix} -4 \\ 3 \end{pmatrix} \tag{3}$$

Therefore, the normal vector of the desired line is given by,

$$\mathbf{n} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \tag{4}$$

$$c = \mathbf{n}^\top \mathbf{A} = \begin{pmatrix} 3 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 6 \end{pmatrix} = 24 \tag{5}$$

Theoretical Solution

Let **D** be the foot of the perpendicular from **B** to the line

$$\begin{pmatrix} \mathbf{m} & \mathbf{n} \end{pmatrix}^\top \mathbf{D} = \begin{pmatrix} \mathbf{m}^\top \mathbf{B} \\ c \end{pmatrix} \tag{6}$$

$$\begin{pmatrix} -4 & 3 \\ 3 & 4 \end{pmatrix} \mathbf{D} = \begin{pmatrix} \mathbf{m}^\top \mathbf{B} \\ 24 \end{pmatrix} \tag{7}$$

$$\mathbf{m}^\top \mathbf{B} = \begin{pmatrix} -4 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0 \tag{8}$$

$$\implies \begin{pmatrix} -4 & 3 \\ 3 & 4 \end{pmatrix} \mathbf{D} = \begin{pmatrix} 0 \\ 24 \end{pmatrix} \tag{9}$$

The augmented matrix is given by,

$$\left(\begin{array}{cc|c} -4 & 3 & 0 \\ 3 & 3 & 24 \end{array}\right) \tag{10}$$

Theoretical Solution

$$R_1 \to \frac{-1}{4}R_1 \implies \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 3 & 3 & 24 \end{array}\right) \tag{11}$$

$$R_2 \to R_2 - 3R_1 \implies \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 0 & 25/4 & 24 \end{array}\right) \tag{12}$$

$$R_2 \to \frac{4}{25}R_2 \implies \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 0 & 1 & 96/25 \end{array}\right) \tag{13}$$

$$R_1 \to R_1 + \frac{3}{4}R_2 \implies \left(\begin{array}{cc|c} 1 & 0 & 72/25 \\ 0 & 1 & 96/25 \end{array}\right) \tag{14}$$

$$\mathbf{D} = \begin{pmatrix} 72/25 \\ 96/25 \end{pmatrix} \tag{15}$$

Theoretical Solution

To find the circle through $\mathbf{B}, \mathbf{C}, \mathbf{D}$ the parameters are given by

$$\begin{pmatrix} 2\mathbf{B} & 2\mathbf{C} & 2\mathbf{D} \\ 1 & 1 & 1 \end{pmatrix}^\top \begin{pmatrix} \mathbf{u} \\ f \end{pmatrix} = -\begin{pmatrix} \|\mathbf{B}\|^2 \\ \|\mathbf{C}\|^2 \\ \|\mathbf{D}\|^2 \end{pmatrix} \tag{16}$$

$$\begin{pmatrix} 0 & 0 & 1 \\ 16 & 0 & 1 \\ 144/25 & 192/25 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ f \end{pmatrix} = \begin{pmatrix} 0 \\ -64 \\ -576/25 \end{pmatrix} \tag{17}$$

The augmented matrix is given by,

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 16 & 0 & 1 & -64 \\ 144/25 & 192/25 & 1 & -576/25 \end{array}\right) \tag{18}$$

$$R_1 \leftrightarrow R_2 \implies \left(\begin{array}{ccc|c} 16 & 0 & 1 & -64 \\ 0 & 0 & 1 & 0 \\ 144/25 & 192/25 & 1 & -576/25 \end{array}\right) \tag{19}$$

Theoretical Solution

$$R_3 \to R_3 - \frac{144}{25}R_1 \implies \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 192/25 & 16/25 & -504/25 \end{array}\right) \tag{21}$$

$$R_2 \leftrightarrow R_3 \implies \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 192/25 & 16/25 & -504/25 \\ 0 & 0 & 1 & 0 \end{array}\right) \tag{22}$$

$$R_2 \to \frac{25}{192}R_2 \implies \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 1 & 1/12 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \tag{23}$$

$$R_2 \to R_2 - \frac{1}{12}R_3 \;,\; R_1 \to R_1 - \frac{1}{16}R_3 \implies \left(\begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \tag{24}$$

Theoretical Solution

Let $\mathbf{n_1}^\top \mathbf{x} = c_1$ and $\mathbf{n_2}^\top \mathbf{x} = c_2$ be the equation of tangents to the circle.

The external point from which the tangents are drawn is $\mathbf{A}$
Compute $\mathbf{\Sigma}$

$$\mathbf{\Sigma} = (\mathbf{Vh} + \mathbf{u})(\mathbf{Vh} + \mathbf{u})^\top - g(\mathbf{h})\mathbf{V} \tag{26}$$

for $\mathbf{h} = \mathbf{A}$

$$g(\mathbf{h}) = \mathbf{h}^\top \mathbf{Vh} + 2\mathbf{u}^\top \mathbf{h} + f = \|\mathbf{h}\|^2 + f = 36 + 0 = 36 \tag{27}$$

$$\mathbf{\Sigma} = (\mathbf{Vh} + \mathbf{u})(\mathbf{Vh} + \mathbf{u})^\top - g(\mathbf{h})\mathbf{V} = \begin{pmatrix} -4 \\ 6 \end{pmatrix} \begin{pmatrix} -4 & 6 \end{pmatrix} - 36 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{28}$$

$$\implies \mathbf{\Sigma} = \begin{pmatrix} -20 & -24 \\ -24 & 0 \end{pmatrix} \tag{29}$$

Theoretical Solution

The eigenvalues of the matrix $\mathbf{\Sigma}$ are given by

$$\left|\mathbf{\Sigma} - \lambda \mathbf{I}\right| = 0 \tag{30}$$

$$\implies \begin{vmatrix} -20-\lambda & -24 \\ -24 & -\lambda \end{vmatrix} \tag{31}$$

$$\implies (-20-\lambda)(-\lambda) - (-24)(-24) = 0 \tag{32}$$

$$\implies \lambda^2 + 20\lambda - 576 = 0 \tag{33}$$

$$\implies \lambda_1 = 16 \text{ and } \lambda_2 = -36 \tag{34}$$

The eigenvectors of the matrix $\mathbf{\Sigma}$ are,

$$\mathbf{p}_1 = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \text{ and } \mathbf{p}_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \tag{35}$$

Theoretical Solution

Define,

$$\mathbf{P} = \begin{pmatrix} \frac{\mathbf{p}_1}{\|\mathbf{p}_1\|} & \frac{\mathbf{p}_2}{\|\mathbf{p}_2\|} \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{13}} & \frac{3}{\sqrt{13}} \\ \frac{-3}{\sqrt{13}} & \frac{2}{\sqrt{13}} \end{pmatrix} \tag{36}$$

The direction vectors of the two tangents are,

$$\mathbf{m} = \mathbf{P} \begin{pmatrix} \sqrt{|\lambda_2|} \\ \pm\sqrt{|\lambda_1|} \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{13}} & \frac{3}{\sqrt{13}} \\ \frac{-3}{\sqrt{13}} & \frac{2}{\sqrt{13}} \end{pmatrix} \begin{pmatrix} 6 \\ \pm 4 \end{pmatrix} \tag{37}$$

$$\implies \mathbf{m}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ and } \mathbf{m}_2 = \begin{pmatrix} 12 \\ -5 \end{pmatrix} \tag{38}$$

$$\implies \mathbf{n}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \mathbf{n}_2 = \begin{pmatrix} 5 \\ 12 \end{pmatrix} \tag{39}$$

Theoretical Solution

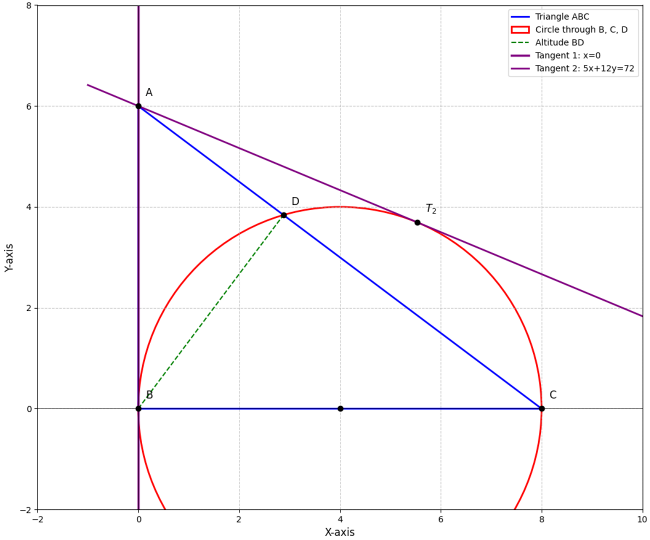
Also,

$$c_1 = \mathbf{n_1}^\top \mathbf{A} \text{ and } c_2 = \mathbf{n_2}^\top \mathbf{A} \tag{40}$$

$$c_1 = 0 \text{ and } c_2 = 72 \tag{41}$$

Therefore, the line $\begin{pmatrix} 5 & 12 \end{pmatrix} \mathbf{x} = 72$ and the $\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 0$ are the two tangents.

Plot



Triangle ABC
Circle through B, C, D
Altitude BD
Tangent 1: x=0
Tangent 2: 5x+12y=72
A
D
T_2
B
C
Y-axis
X-axis
8
6
4
2
0
−2
−2
0
2
4
6
8
10